BS-Nets: An End-to-End Framework for Band Selection of Hyperspectral Image

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***Abstract*— Hyperspectral image (HSI) consists of hundreds of continuous narrowbands with high spectral correlation, which would lead to the so-called Hughes phenomenon and the high computational cost in processing. Band selection (BS) has been proven to be effective in avoiding such problems by removing redundant bands. However, many existing BS methods separately estimate the significance for every single band and cannot fully consider the nonlinear and global interaction between spectral bands. In this article, by assuming that a complete HSI band set can be reconstructed from its few informative bands, we propose a unified BS framework, BS Network (BS-Net). The framework consists of a band attention module (BAM), which aims to explicitly model the nonlinear interdependences between spectral bands, and a reconstruction network (RecNet), which is used to restore the original HSI from the learned informative bands, resulting in a flexible architecture. The resulting framework is end-to-end trainable, making it easier to train from scratch and to combine with many existing networks. We implement two versions of BS-Nets, respectively, using fully connected networks (BS-Net-FC) and convolutional neural networks (BS-Net-Conv), and extensively compare their results with popular existing BS approaches on three real hyperspectral data sets, showing that the proposed BS-Nets can accurately select informative band subset with less redundancy and outperform the competitors in terms of classification accuracy with competitive time cost.**

***Index Terms*— Attention mechanism, band selection (BS), deep neural networks (DNNs), hyperspectral image (HSI), spectral reconstruction.**

I. Introduction

HYPERSPECTRAL images (HSIs) acquired by remote sensors consist of hundreds of narrowbands containing rich spectral and spatial information, which provides an ability to accurately recognize the region of interest. Over the past decade, HSIs have been widely applied in various fields, ranging from agriculture [1] and land management [2] to medical imaging [3] and forensics [4].

As the development of hyperspectral imaging techniques, the spectral resolution has been improved greatly, resulting in difficulty of analyzing. According to the characteristic of hyperspectral imaging, there is a high correlation between the adjacent spectral bands [5]–[7]. The high-dimensional HSI data not only increase the time complexity and space complexity but also lead to the so-called Hughes phenomenon or curse of dimensionality [8], [9]. As a result, redundancy reduction becomes particularly important for HSI processing.

Band selection (BS) [10]–[12], also known as feature selection, is an effective redundancy reduction scheme. Its basic idea is to select a significant band subset that includes most information of the original band set. In contrast to the feature extraction methods [13] which reduces dimensionality based on the complex feature transformation, BS keeps the main physical property containing in HSIs [5], which makes it easier to explain and apply in practice.

BS methods basically can be classed as supervised and unsupervised methods [14] based on whether the prior knowledge is used. Due to more robust performance and higher application prospect, an unsupervised BS method has attracted a great deal of attention over the last few decades. Unsupervised BS methods can be further divided into three categories: searching-, clustering-, and ranking-based methods [15]. The searching-based BS methods treat BS as a combinational optimization problem and optimize it using a heuristic searching method [16], such as multiobjective optimization-based BS (MOBS) [5], [17], [18]. However, heuristic searching methods are generally time-consuming [19]–[22]. The clustering-based BS methods assume that spectral bands are clusterable [14], [23]. Since the similarity between spectral bands is made full consideration, clustering-based methods have achieved great success in recent years, for example, subspace clustering (ISSC) [7], [11] and sparse nonnegative matrix factorization clustering (SNMF) [24]. The ranking-based BS methods endeavor to assign a rank or weight for each spectral band by estimating each band's significance, e.g., maximum-variance principal component analysis (MVPCA) [25], sparse representation (SpaBS) [26], [27], and geometry-based BS (OPBS) [28]. In many probability models, such as Bayes methods [29], [30], the ranking strategy is popular for feature weighing.

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Nevertheless, many existing BS methods are based on the linear transformation of spectral bands, resulting in the lack of consideration of the inherent nonlinear relationship between the spectral bands. Furthermore, most of the BS methods commonly view every single spectral band as a separate image or point and evaluate its significance independently. For example, clustering-based BS methods are essentially clustering spectral images with single channel [7], [11], [26]. Therefore, these methods cannot take the global spectral interrelationship into account and are difficult to combine with the subsequent processing, such as classification [31].

Recently, deep neural network (DNN) [32], [33] has achieved great success in HSI processing ranging from HSI classification [9], [34] and feature extraction [35], [36] apart from BS, due to its powerful ability to learn hierarchical and nonlinear features in an end-to-end trainable way. According to the universal approximation theorem, DNN can be used to approximate any continuous function. This has inspired a large number of interesting and promising neural network models and architectures, for example, attention mechanism [37], which is designed to approximate a certain relationship between contexts of inputs. Equipped with attention mechanism, DNNs are enabled to focus on the salient components of inputs. In the computer vision community, the characteristic of the attention mechanism has been widely applied for channel weighting [38]–[40] and pixel weighing [41], [42] for many visual tasks. However, there is no unified framework based on the attention module for HSI BS.

To take advantage of the attention mechanism in HSI BS, in this article, we develop a unified BS network (BS-Net) framework that treats HSI BS as a spectral reconstruction task. Specifically, we assume that the complete spectral band set can be sparsely reconstructed from an informative band subset. The framework is achieved by a self-supervised network architecture equipped with a band attention module (BAM). Instead of estimating each band independently, BS-Net allows to input full bands 2-D spectral samples or 3-D spectral–spatial samples and explicitly models the global nonlinear interdependences between the spectral bands. For the flexible implementation and end-to-end trainable structure, BS-Net can be viewed as a unified BS framework.

To sum up, the main contributions of this article are as follows.

1) By assuming that the spectral band set is reconstructable from an informative subset, an end-to-end framework is developed for HSI BS, which explicitly models the nonlinear interdependences between the spectral bands. The successful attempt signifies that considering nonlinear relations between the bands is important for BS, which offers an alternative orientation for unsupervised BS. To the best of our knowledge, this is the first unified DNN-based framework for unsupervised HSI BS.
2) According to different application scenarios, two different versions of BS-Nets, named BS-Net-FC and BS-Net-Conv, are implemented, which are characterized by implementation using fully connected networks and with convolutional networks, respectively, which enables BS-Net to directly select significant bands from 2-D spectral inputs or 3-D spectral–spatial data. As a result, BS-Net can be integrated with many existing neural networks as a preprocessing step of the subsequent HSI classification.
3) The proposed BS-Nets framework achieves the state-of-the-art results on three real HSI data sets. Especially, we propose to analyze the selected band subsets from not only the classification performance but also the bands' distribution and the quantization of entropy and mean spectral divergence (MSD).

The rest of this article is structured as follows. We first define the notations, review the basic concepts of deep learning and attention mechanism, and describe our motivation in Section II. Second, we introduce the proposed BS-Net framework and its two versions of implementation in Section III. Next, in Section IV, we explain the experiments that performed to investigate the performance of the proposed methods, compare with many existing BS methods, and discuss their results. Finally, we conclude with a summary and final remarks in Section V.

II. Preliminary

A. Definition and Notations

We denote a 3-D HSI cube consisting of b spectral bands and $N \times M$ pixels as $\boldsymbol{I} \in \mathbb{R}^{N \times M \times b}$. For convenience, we regard $\boldsymbol{I}$ as a set $\boldsymbol{B} = \{\boldsymbol{B}_i\}_{i=1}^{b}$ that contains b band images, where $\boldsymbol{B}_i$ indicates the ith band image. HSI BS can thus be formally defined as a function $\psi : \Omega = \psi(\boldsymbol{B})$ that takes all bands as the input and produces a band subset with as less as possible reduction of redundant information and satisfied $\Omega \subseteq \boldsymbol{B}$, $|\Omega| = k < b$.

In the following, unless as otherwise specified herein, we uniformly use tensors to represent the inputs, outputs, and intermediate outputs involved in the neural networks. For example, the input of a convolutional layer is denoted as a 3-D tensor $\boldsymbol{x} \in \mathbb{R}^{n \times m \times c}$, where $n \times m$ is the spatial size of the input feature maps and c is the number of the channels.

B. Convolutional Neural Networks

Deep learning has achieved great success in numerous applications ranging from image recognition to natural language processing [32], [43], [44]. The collection of deep learning methods includes convolutional neural networks (CNNs) [33], [45], generative adversarial networks (GANs) [46], [47], and recurrent neural networks (RNNs) [48], to name a few. In this section, we take CNN as an example to introduce the basic idea of DNNs since it is the most popular deep-learning method in HSI processing.

CNNs are inspired by the natural visual perception mechanism of the living creatures [33]. The classical CNN consists of multiple layers of convolutional operations with nonlinear activations, sometimes, followed by a regression layer. A schematic of the basic CNN architecture is shown in Fig. 1. We define CNN as a function that takes a tensor $\boldsymbol{x} \in \mathbb{R}^{m \times n \times c}$ as the input and produces a certain output $\boldsymbol{y}$. The function can be written as $\boldsymbol{y} = \boldsymbol{f}(\boldsymbol{x}; \Theta)$, where Θ is the trainable parameters consisting of weights and biases involved in CNN.

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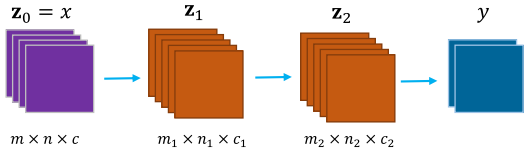


Fig. 1. Example of classical CNN with input $\boldsymbol{x}$, output y, and two layers of convolutional feature maps z_1 and z_2.

The training of CNN includes two stages. The first stage is values feedforward in which each layer yields a dozen of feature maps $\boldsymbol{h}_i \in \mathbb{R}^{m_i \times n_i \times c_i}$. Let $\hbar : \mathbb{R}^{m \times n \times c} \to \mathbb{R}^{m_i \times n_i \times c_i}$ be the convolutional operation and $\boldsymbol{\sigma}$ be an elementwise nonlinear function, such as sigmoid and rectified linear unit (ReLU). The convolutional layer can be represented as

$$\boldsymbol{h}_i = \boldsymbol{\sigma}(\hbar(\boldsymbol{x}; \mathbf{W}) + \boldsymbol{b}) \tag{1}$$

where $\mathbf{W}$ and $\boldsymbol{b}$ indicate weights (also known as convolutional kernels or convolutional filters) and bias, respectively.

The second stage is called error backpropagation that updates the parameters using the gradient descent method. The ultimate goal of CNN is to find an appropriate group of filters to minimize the cost function, e.g., mean square error (mse) function. The cost function can be denoted as

$$\mathcal{J}(\Theta) = \text{Cost}(\boldsymbol{y}, \boldsymbol{f}(\boldsymbol{x}; \Theta)). \tag{2}$$

The parameters updating is given by

$$\Theta =: \Theta - \eta \frac{\partial \mathcal{J}}{\partial \Theta} \tag{3}$$

where η is the learning rate (or step size), and the partial derivatives of the cost function with respect to the trainable parameters can be calculated using the chain rule.

C. Attention Mechanism

Attention is, to some extent, motivated by how human pays attention to different regions of an image or correlates words in one sentence. Attention mechanism is widely applied across a range of tasks, including image processing [38]–[40] and natural language processing [37], [49], [50]. In this article, we focus mainly on the attention mechanism involved in computer vision. In [38], the attention in visual tasks is defined as a method to bias the allocation of available processing resources toward the most informative components of an input signal. Mathematically, an attention module can be expressed as follows:

$$\boldsymbol{a} = \boldsymbol{f}_{\Theta}(\boldsymbol{x}) \tag{4}$$

where $\boldsymbol{a}$ is called as attention map that is formulated by an attention module $\boldsymbol{f}$ parameterized by Θ. Generally, $\boldsymbol{f}$ contains a gating function, e.g., softmax, which is used to scale the attention map. In practice, $\boldsymbol{f}$ is widely achieved by using neural networks, resulting in a multiple-branch structure. To link up the attention map and a feature map $\boldsymbol{z}$ which is learned from $\boldsymbol{x}$, an elementwise operation, $\boldsymbol{h} = \boldsymbol{a} \otimes \boldsymbol{z}$, could be used in a complete attention method. Therefore, the resulted feature map $\boldsymbol{h}$ will pay more attention to the part of interest.

According to the different concern, attention modules can basically be divided into three categories, i.e., spatial attention, channel attention, and joint attention. Their representative sketches are shown in Fig. 2. The spatial attention is used to learn the relationship between one pixel and all other positions. As shown in Fig. 2(a), a spatial attention map is generally represented as a matrix with the same size as the input spatial domain. For instance, the nonlocal network [51] uses spatial attention to learn an affinity (pairwise) matrix that captures the long-range dependences of the input signals. Similarly, from Fig. 2(b), channel attention aims to model the channelwise relationship. In a squeeze-and-excitation network [38], channel attention is designed as a simple network branch but performs dramatically and can easily combine with any existing network. Joint attention is given in Fig. 2(c), where spatial attention and channel attention are combined to model the different specific relationships. It has been proven to be powerful for image classification [42], [52], scene segmentation [52], image captioning [41], and so on.

As mentioned earlier, the attention mechanism has the ability to select salient features. For the task of HSI BS, we expect to learn a band weight vector that can explicitly reflect the relationship between the spectral bands. From this point of view, BS actually can be interpreted as a channel attention module. In this article, we refer to the attention used for HSI BS as band attention. In Section III, we will design a different type of BAMs to select significant spectral band.

D. Motivation

This article is based on the assumption, i.e., a complete spectral band set can be sparsely reconstructed from its informative subset. Unfortunately, it is NP-hard to determine the informative band subset. An available way to do this is to use the self-supervised scheme, such as sparse self-expressive model [15], [53]. By assuming that the data are linear, self-expressive can be written as $\mathbf{XC} = \mathbf{X}$, where $\mathbf{X} \in \mathbb{R}^{MN \times b}$ and $\mathbf{C} \in \mathbb{R}^{b \times b}$ is a coefficient matrix or affinity matrix subjecting to minimum nonzero elements. In order to make the linear representation more robust, structured information can be considered in linear models [54], [55]. Since the affinity matrix $\mathbf{C}$ reveals the intrinsic relationship between spectral bands, it can be used for BS and band clustering. However, it is essentially a linear sparse representation model and thus will face the following shortcomings. 1) The linear assumption is generally difficult to meet in complex HSI data. 2) The affinity matrix $\mathbf{C}$ cannot directly reflect the contribution of each band during the reconstruction. To select significant bands, additional processes are required. 3) Sparse representation utilities shallow spectral features, and thus, it cannot consider the high-level semantic information. Hence, this article attempts to utilize self-supervised DNNs to explicitly model the interdependence between the spectral bands by considering spectral–spatial information.

III. BS-Nets

In this section, we first introduce the main components included in the BS-Nets general architecture. Then, we give

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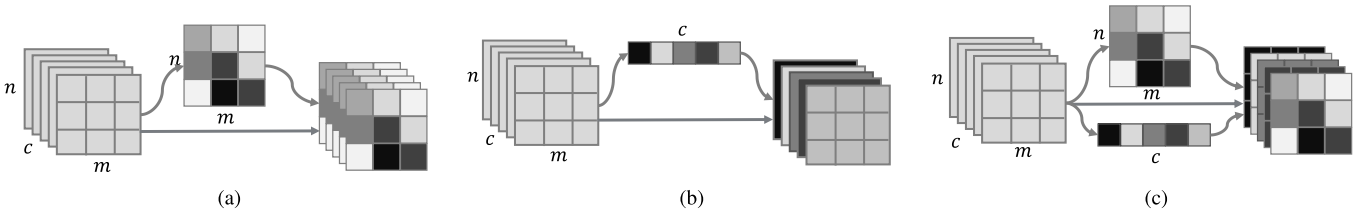


Fig. 2. Representative sketches of (a) spatial attention, (b) channel attention, and (c) joint attention in the context of computer vision.

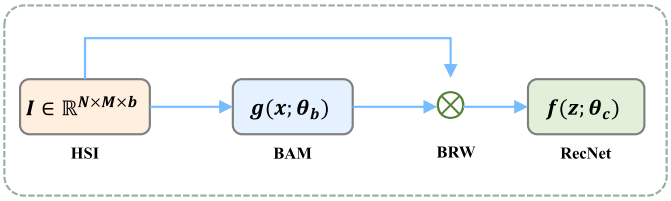


Fig. 3. Overview of BS-Nets. A given HSI data are first passed onto a BAM to explicitly model the nonlinear interdependences between the spectral bands. Then, the input HSI is reweighted bandwisely using a band reweighting (BRW) operation. Finally, a RecNet is adopted to restore the original spectral bands from the reweighted ones.

two versions of implementations of the BS-Nets based on fully connected networks and CNNs, respectively. Finally, we show a discussion on the BS-Nets.

A. Architecture of BS-Nets

The key to the BS-Nets is to convert the BS as a sparse band reconstruction task, i.e., recover the complete spectral information using a few informative bands. For a given spectral band, if it is informative, then it will be essential for a spectral reconstruction. To this end, we design a DNN based on the attention mechanism. In Fig. 3, we show the overall architecture of the proposed framework, which consists of three components, BAM, BRW, and reconstruction network (RecNet). The detailed introduction is given as follows.

The BAM is a branch network that we use to learn the band weights. As shown in Fig. 3, BAM directly takes HSI as the input and aims to fully extract the interdependences between the spectral bands. We express BAM as a function g that takes a certain HSI cube $\boldsymbol{x}$ as the input and produces a nonnegative band weights' tensor, $\boldsymbol{w} \in \mathbb{R}^{1\times 1\times b}$

$$\boldsymbol{w} = g(\boldsymbol{x}; \boldsymbol{\Theta}_b) \tag{5}$$

where $\boldsymbol{\Theta}_b$ denotes the trainable parameters involved in the BAM. To guarantee the nonnegativity of the learned weights, Sigmoid function is adopted as the activation of the output layer in BAM, which is written as

$$\phi(\boldsymbol{w}) = \frac{1}{1+e^{-\boldsymbol{w}}}. \tag{6}$$

To create an interaction between the original inputs and their weights, a bandwise multiplication operation is conducted. We refer to this operation as BRW. It can be explicitly represented as follows:

$$\mathbf{z} = \boldsymbol{x} \otimes \boldsymbol{w} \tag{7}$$

where $\otimes$ indicates the bandwise production between $\boldsymbol{x}$ and $\boldsymbol{w}$, and $\mathbf{z}$ is the reweighted counterpart of the input $\boldsymbol{x}$.

In the next step, we employ the RecNet to recover the original spectral band from the reweighted counterpart. Similarly, we define the RecNet as a function f that takes a reweighted tensor $\mathbf{z}$ as the input and outputs its prediction

$$\hat{\boldsymbol{x}} = f(\mathbf{z}; \boldsymbol{\Theta}_c) \tag{8}$$

where $\hat{\boldsymbol{x}}$ is the prediction output for the original input $\boldsymbol{x}$, and $\boldsymbol{\Theta}_c$ denotes the trainable parameters involved in RecNet.

In order to measure the reconstruction performance, we use the mse as the cost function, denoted as $\mathcal{L}$. We define it as follows:

$$\mathcal{L} = \frac{1}{2S}\sum_{i=1}^{S} \|x_i - \hat{x}_i\|_2^2 \tag{9}$$

where S is the number of training samples. Moreover, we desire to keep the band weights as sparse as possible such that we can interpret them more easily. For this purpose, we impose an L_1 norm constraint on the band weights. The resulting loss function is given as follows:

$$\mathcal{L}(\Theta_b, \Theta_c) = \frac{1}{2S}\sum_{i=1}^{S} \|x_i - \hat{x}_i\|_2^2 + \lambda \sum_{i=1}^{S} \|w_i\|_1 \tag{10}$$

where λ is a regularization coefficient that balances the minimization between the reconstruction error and the regularization term. Equation (10) can be optimized by using a gradient descent method, such as stochastic gradient descent (SGD) and adaptive moment estimation (Adam).

According to the learned sparse band weights, we can determine the informative bands by averaging the band weights for all the training samples. The average weight of the jth band is computed as

$$\bar{w}_j = \frac{1}{S}\sum_{i=1}^{S} w_{ij}. \tag{11}$$

Those bands that have larger average weights are considered to be significant since they make more contributions to the reconstruction. In practice, the top k bands are selected as the significant band subset. The pseudocode of BS-Nets is given in Algorithm 1.

B. BS-Net-FC

In Fig. 4(a), we show the first implementation of BS-Net based on fully modeling the nonlinear relationship between the spectral information. In this case, both BAM and RecNet are

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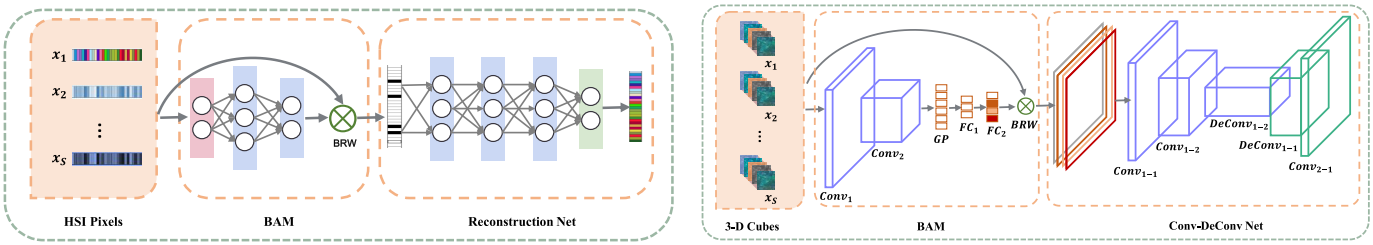


Fig. 4. Implementation details of BS-Nets based on different networks. (a) BS-Net based on fully connected networks with spectral inputs. (b) BS-Net based on CNNs with spectral–spatial inputs.

Algorithm 1 Pseudocode of BS-Nets

Input: HSI cube: $\boldsymbol{I} \in \mathbb{R}^{N\times M\times b}$; Band subset size: k; and BS-Nets hyper-parameters.
Output: Informative band subset.

```
Preprocess HSI and generate training samples;
Random initialize Θ_b and Θ_c according to the given network configure;
while Model is convergent or maximum iteration is met do
    Sample a batch of training samples x;
    Calculate bands weights: w = g(x; Θ_b);
    Re-weight spectral bands: z = x ⊗ w;
    Reconstruct spectral bands: x̂ = f(z; Θ_c);
    Update Θ_b and Θ_c by minimizing Eq.(10) using Adam algorithm;
end
Calculate average band weights according to Eq. (11);
Select top k bands;
```

implemented with fully connected networks, and thus, we refer to this BS-Net as BS-Net-FC.

As shown in Fig. 4(a), the BAM is designed as a bottleneck structure with multiple fully connected layers, with ReLu activations for all the middle hidden layers. According to the information bottleneck theory [56], bottleneck structure would be favorable for the extraction of information, although different structures are allowed in BS-Nets.

In BS-Net-FC, we use spectral vectors (pixels) as the training samples. For convenience, we denote the training set comprising S samples as a 4-D tensor $\mathbf{X} \in \mathbb{R}^{S\times 1\times 1\times b}$, where $S = M \times N$. By rewriting the band weights in the tensor form, represented as $\mathbf{W} \in \mathbb{R}^{S\times 1\times 1\times b}$, the BRW is actually an elementwise production operation that can be written as $\mathbf{Z} = \mathbf{X} \otimes \mathbf{W}$, where $\mathbf{Z}$ is the reweighted spectral inputs. In RecNet, we use a simple multilayer perceptron model with the same number of hidden neurons with ReLu activations to reconstruct spectral information.

C. BS-Net-Conv

During the training in BS-Net-FC, only the spectral information is considered. The lack of consideration for the spatial information would result in low-efficiency use of the spectral–spatial information containing in HSI. To enhance the BS-Net-FC, we implement the second BS-Net by using convolutional networks, which is termed BS-Net-Conv. The schematic of the implementation is given in Fig. 4(b).

In the BAM, we first employ several 2-D convolutional layers to extract spectral and spatial information simultaneously. Then, a global pooling (GP) layer is used to reduce the spatial size of the resulting feature maps. Finally, the final band weights $\mathbf{W}$ are generated by a few fully connected layer and used to reweight the spectral bands. BS-Net-Conv adopts a convolutional–deconvolutional network (Conv–DeConv Net) to implement the RecNet. Similar to the classical autoencoder, Conv–DeConv Net includes a convolutional encoder that extracts deep features and a deconvolutional decoder that up-samples feature maps.

Instead of using single pixels, BS-Net-Conv takes 3-D HSI patches, which includes spectral and spatial information as the training samples. To generate enough training samples, we use a rectangular window of size $a \times a$ to slides across the given HSI with stride t. The generated training samples can be denoted as $\mathbf{X} \in \mathbb{R}^{S\times a\times a\times b}$, where $S = (M-a)/t \times (N-a)/t + 1$. Notice that the number of training samples in BS-Net-Conv is less than that in BS-Net-FC.

D. Remarks on BS-Nets

The key to our proposed BS-Net framework is to use DNNs to explicitly learn spectral bands weights. Compared with the existing BS methods, the framework has the following advantages. The first is the framework that is end-to-end trainable, making it easy to combine with specific tasks and existed neural networks, such as deep learning-based HSI classification. The second is the framework that is capable of adaptively exacting spectral and spatial information, which avoids hand-designed features and reduces the noise effect. The third is the framework that is nonlinear, enabling it to make full exploration of the nonlinear relationship between the bands. The fourth is the framework that is flexible to be implemented with diverse networks.

The main difference between BS-Net-FC and BS-Net-Conv is that: 1) BS-Net-FC models the interdependence between the bands from 2-D spectral information, whereas BS-Net-Conv does this from 3-D spectral-spatial information and 2) for the reason mentioned 1), BS-Net-FC is achieved by fully connected networks, while BS-Net-Conv is achieved by CNNs.

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TABLE I
SUMMARY OF INDIAN PINES, PAVIA UNIVERSITY, AND SALINAS DATA SETS

Data sets	Indina Pines	Pavia University	Salinas
Pixels	145×145	610×340	512×217
Channels	200	103	204
Classes	16	9	16
Labeled pixels	10249	42776	54129
Sensor	AVIRIS	ROSIS	AVIRIS

TABLE II
HYPERPARAMETERS SETTINGS FOR DIFFERENT BS METHODS

Baselines	Hyper-parameters
ISSC	$\lambda = 1e5$
SpaBS	$\lambda = 1e2$
MVPCA	–
SNMF	$maxiter = 100$
MOBS	$maxiter = 100, NP = 100$
OPBS	–
BS-Net-FC	$\lambda = 1e-2, \eta = 2e-3, maxiter = 100$
BS-Net-Conv	$\lambda = 1e-2, \eta = 2e-3, maxiter = 100$

TABLE III
CONFIGURATION OF BS-NET-FC FOR INDIAN PINES DATA SET

Branch	Layer	Hidden Neurons	Activation
BAM	Input	200	–
	FC1-1	64	ReLU
	FC1-2	128	ReLU
	FC1-3	200	Sigmoid
RecNet	FC2-1	64	ReLU
	FC2-2	128	ReLU
	FC2-3	256	ReLU
	FC2-4	200	Sigmoid

TABLE IV
CONFIGURATION OF BS-NET-CONV FOR INDIAN PINES DATA SET

Branch	Layer	Kernel	Activation
BAM	Conv1	$3 \times 3 \times 64$	ReLU
	GP	–	–
	FC1	128	ReLU
	FC2	200	Sigmoid
RecNet	Conv1-1	$3 \times 3 \times 128$	ReLU
	Conv1-2	$3 \times 3 \times 64$	ReLU
	DeConv1-2	$3 \times 3 \times 64$	ReLU
	DeConv1-1	$3 \times 3 \times 128$	ReLU
	Conv2-1	$1 \times 1 \times 200$	Sigmoid

IV. RESULTS

A. Setup

In this section, we will widely evaluate the performance of the proposed BS-Nets on three real HSI data sets: Indian Pines, Pavia University, and Salinas. The summary of the three data sets is shown in Table I. For each data set, we first investigate the model convergence by analyzing the training loss, classification accuracy, and band weights. Next, we compare the classification performance for BS-Nets with many existing BS methods, i.e., ISSC [11], SpaBS [26], MVPCA [25], SNMF [24], MOBS [5], and OPBS [28]. The hyperparameters settings of all the methods are listed in Table II. To better demonstrate the performance, we also compare with all bands. Finally, we make a deep analysis on the selected band subsets from aspects of visualization and quantification.

Similar to the evaluation strategy adopted in [5] and [28], we use support vector machine (SVM) [8] with radial basis function kernel as the classifier to evaluate the classification performance of the selected band subsets. For the sake of fairness, we randomly select 5% of labeled samples from each data set for the training set and the rest for the testing set. Three popular quantitative indices, i.e., overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa), are calculated by evaluating each BS method for 20 independent runs.

To quantitatively analyze the selected band subsets, entropy and MSD [5], [57] of band subsets are calculated. For a single band $\boldsymbol{B}_i$, its entropy is defined as follows:

$$H(B_i) = -\sum_{y \in \Psi} p(y) \log(p(y)) \tag{12}$$

where y denotes a gray level of the histogram of the ith band $\boldsymbol{B}_i$ and $p(y) = n(y)/(N \times M)$ indicates the ratio (probability) of the number of y to that all pixels. According to the characteristic of entropy, the larger the entropy is, the more image details the band contains [5].

MSD is an average measurement index for a band subset $\boldsymbol{B}$, which is expressed as

$$M(B) = \frac{2}{k(k-1)} \sum_{i=1}^{k} \sum_{j=1}^{k} D_{\mathrm{SKL}}(B_i \| B_j) \tag{13}$$

where D_{SKL} is the symmetrical Kullback–Leibler divergence that measures the dissimilarity between $\boldsymbol{B}_i$ and $\boldsymbol{B}_j$. Specifically, D_{SKL} is defined as follows:

$$D_{\mathrm{SKL}}(\boldsymbol{B}_i \parallel \boldsymbol{B}_j) = D_{\mathrm{KL}}(\boldsymbol{B}_i \parallel \boldsymbol{B}_j) + D_{\mathrm{KL}}(\boldsymbol{B}_j \parallel \boldsymbol{B}_i) \tag{14}$$

where $D_{\mathrm{KL}}(\boldsymbol{B}_i \parallel \boldsymbol{B}_j)$ can be computed from the gray histogram information. From (13), MSD evaluates the redundancy among the selected bands, that is, the larger the value of the MSD is, the less redundancy is contained among the selected bands.

The configuration of BS-Nets implemented in our experiments are shown in Tables III and IV. In reprocessing, we scale all the HSI pixel values to the range $[0, 1]$. All the baseline methods are evaluated with Python 3.5 running on an Intel Xeon E5-2620 2.10-GHz CPU with 32-GB RAM. In addition, we implement BS-Nets with TensorFlow-GPU 1.6[1] and accelerate them on an NVIDIA TITAN XP GPU with 11-GB graphics memory. One may refer to https://github.com/AngryCaihttps://github.com/AngryCai for the source codes and trained models.

B. Results on Indian Pines Data Set

1) Data Set: This scene was gathered by the AVIRIS sensor over the Indian Pines test site in Northwestern Indiana and consists of 145 × 145 pixels and 224 spectral reflectance bands in the wavelength range of 0.4–2.5 ($\times 10^{-6}$) m.

[1] https://tensorflow.google.cn

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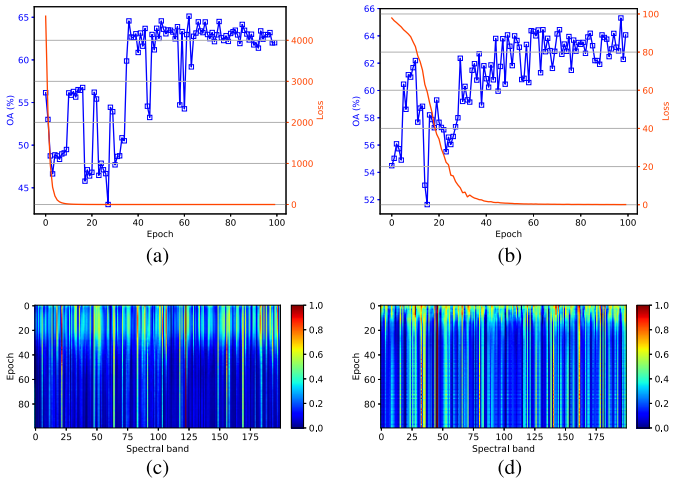


Fig. 5. Analysis of the convergence of BS-Nets on the Indian Pines data set. Visualization of loss versus accuracy under different iterations for (a) BS-Net-FC and (b) BS-Net-Conv. Visualization of average band weights under varying iterations for (c) BS-Net-FC and (d) BS-Net-Conv.

The scene contains two-thirds agriculture, and one-third forest or other natural perennial vegetation. There are two major dual-lane highways, a rail line-as well as some low-density housing and other built structures-and smaller roads. Since the scene is taken in the month of June, some of the crops, corn and soybeans, are in early stages of growth with less than 5% coverage. The ground-truth available is designated into 16 classes and is not all mutually exclusive. We have also reduced the number of bands to 200 by removing bands covering the region of water absorption: [104–108], [150–163], and 220.

2) Analysis of Convergence of BS-Nets : To analyze the convergence of BS-Nets, we train BS-Nets for 100 iterations and plot their training loss curves and classification accuracy using the best five significant bands. The results of BS-Net-FC and BS-Net-Conv are shown in Fig. 5(a) and (b), respectively. Observing from Fig. 5(a) and (b), the reconstruction errors decrease with iterations, and at the same time, the classification accuracies increase. The loss values of BS-Net-FC are very close to zero after 20 iterations, and the classification accuracy finally stabilizes around 63% after 40 interactions. A similar trend can be found from BS-Net-Conv, showing that the proposed BS-Nets are easy to train with high convergence speed. Furthermore, we can find that the classification accuracy is increased from 42% to 63% for BS-Net-FC and 52% to 64% for BS-Net-Conv. Therefore, the well-optimized BS-Nets can, respectively, achieve 21% and 12% improvement in terms of classification accuracy, which demonstrates the effectiveness of the proposed BS-Nets.

We further visualize the average band weights obtained by BS-Net-FC and BS-Net-Conv with different iterations in Fig. 5(c) and (d), respectively. To better show the change trend of band weights, we scale these weights into range [0, 1]. The horizontal and vertical axes represent the band number and iterations, respectively, where each column depicts the change of one band's weights. As we can see, the band weights distribution becomes gradually sparser and sparser. Meanwhile, the informative bands become easier to be distinguished due to the trivial bands will finally be assigned with very small weights.

3) Performance Comparison: To demonstrate the effectiveness of both BS-Nets, we compare the classification performance of different BS methods under different band subset sizes. Three indices, i.e., OA, AA, and Kappa, are computed under band subset sizes ranging from 3 to 30 with two-band interval. All bands' performance is also compared as an important reference. To ensure a reliable classification result, we conduct each method for 20 times and randomly select the training set and test set during each time.

Fig. 6(a)–(c) shows the average comparison results of OA, AA, and Kappa, respectively. From Fig. 6(a)–(c), BS-Net-Conv achieves the best OA, AA, and Kappa when the band subset size is larger than 5, while BS-Net-FC achieves comparable performance with MOBS but is superior to the other five competitors. BS-Net-Conv requires, respectively, 5, 17, and 17 bands to achieve better OA, AA, and Kappa than all bands, while other methods either are worse than all bands or have to use more bands. Furthermore, a counter-intuitive phenomenon that the classification performance is not always increased by selecting more bands can be found from the results. For example, ISSC and SNMF display a significant decreasing trend when the band subset size is over 7. The phenomenon can be explained as the so-called Hughes phenomenon [58], i.e., the classification accuracy increases first and then decreases with the selected bands. However, even if so, the two BS-Nets occurs the phenomenon obviously later than other methods, which means that our methods are able to select more effective band subset. It is worth noting that the best subset size determined by BS-Net-Conv is closed to 25, which is consistent with that evaluated by the virtual dimensionality (VD) analysis [59], [60] using the false alarm probability $p_F = 10^{-5}$. In the following experiments, we will set the band subset size of the Indian Pines data set to be 25.

Fig. 6(d)–(f) shows the classification performance under different percentages of training samples. In this experiment, we fix the band subset to be 25 and make the training size changing from 5% to 30% with 5% interval. The results show that BS-Net-Conv is significantly better than all the other BS methods in terms of OA, AA, and Kappa. Compared with all bands, BS-Net-Conv always achieves better OA under different sizes of training samples, and moreover, when the training size is less than 10%, BS-Net-Conv also yields better AA and Kappa. In addition, BS-Net-FC has comparable performance to ISSC but is better than SpaBS, MVPCA, SNMF, and OPBS.

In Table V, we show the detailed classification performance of different methods by selecting the best 25 bands and using 5% training samples. As can be seen, BS-Net-Conv achieves the best OA (70.61%), AA (75.17%), and Kappa (0.717). For some classes which contain limited training samples, such as No. 7 and No. 9 class, BS-Net-FC and BS-Net-Conv can still yield better classification performance than the other competitors. Compared with all bands, BS-Net-Conv wins 13 classes; moreover, it achieves 7.83%, 1.37%, and 0.017 increase in terms of OA, AA, and Kappa, respectively. BS-Net-FC and MOBS have a similar performance, and both of them

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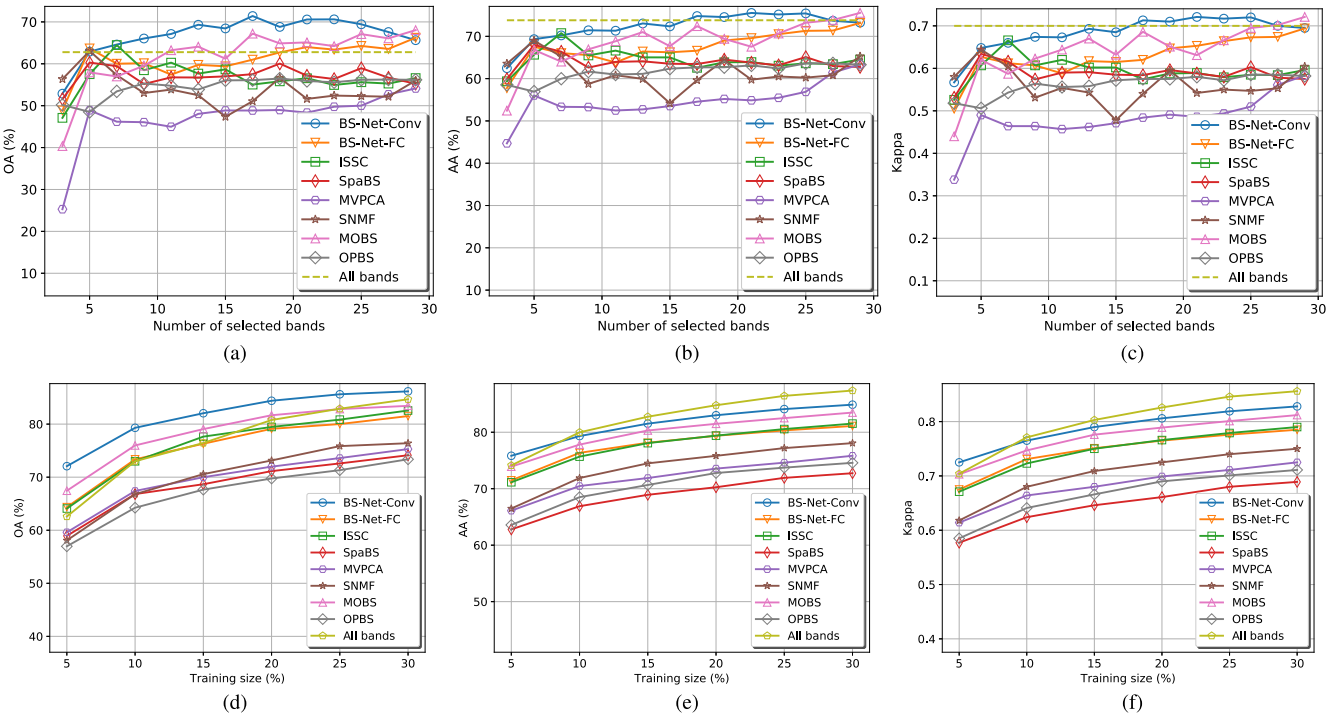


Fig. 6. Performance comparison of different BS methods with different band subset sizes (a)–(c) and different training sample sizes (d)–(e) on the Indian Pines data set. (a) and (d) OA. (b) and (e) AA. (c) and (f) Kappa.

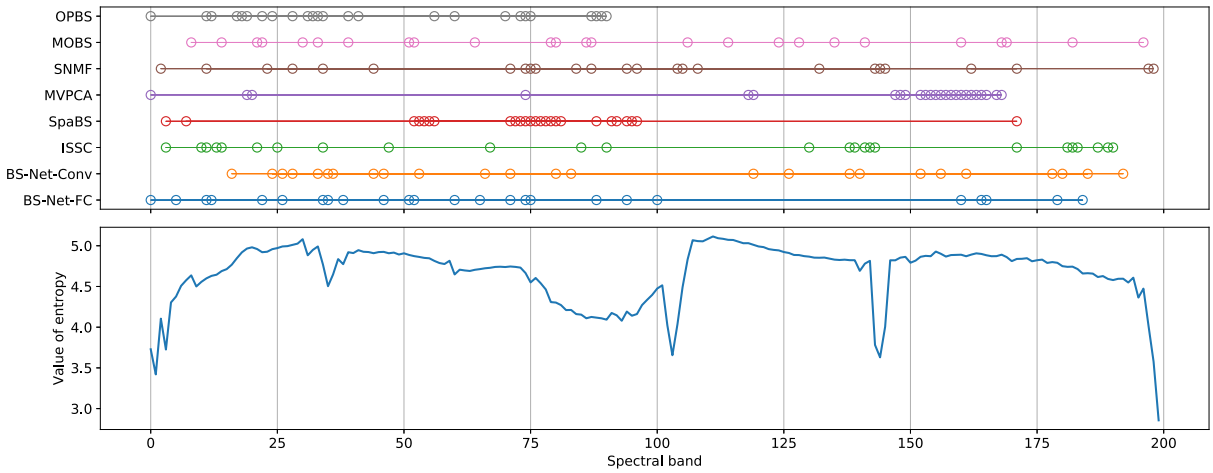


Fig. 7. Best 25 bands of the Indian Pines data set selected by (Top) different BS methods and (Bottom) entropy value of each band.

are superior to all bands. By comparing the results of the two BS-Nets, BS-Net-Conv is significantly better than BS-Net-FC, which demonstrates that spectral–spatial information is more effective for the BS than using only spectral information.

4) Analysis of the Selected Bands: We extensively analyze the selected bands in this section. Fig. 7 shows the best 25 bands [2] of the Indian Pines data set selected by different methods. To better show the band distribution, we also indicate the corresponding specific locations of these bands on the spectrum in Fig. 7 (top). One can find the detailed band indices in Table IX in the Appendix. Each row represents a BS method with its corresponding locations of the selected bands. As can be seen, the results obtained by the two BS-Nets contain less continuous bands and with a relatively uniform distribution. Based on the fact that the adjacent bands generally include high correlation, less redundancy will be contained among the band subsets of BS-Nets. By contrast, there are many continuous bands existing in OPBS, MVPCA, and SpaBS. Thus, their selected band subsets are highly correlated and result in worse classification performance.

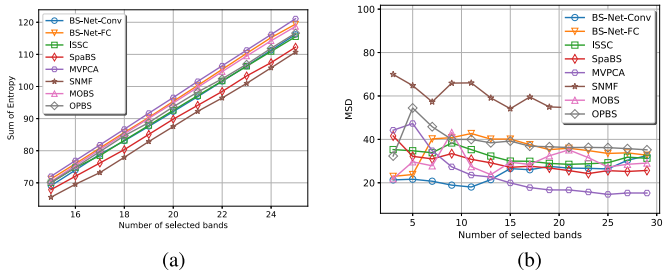
To quantitatively analyze the band subsets, we give entropy of each band in Fig. 7 (bottom). As is well known, entropy reflects how informative a band image is. Those bands with extremely low entropy compared with their adjacent bands can be regarded as noisy bands with little information, i.e., [104, 105], [144, 145], and [198, 199, 200]. It can be seen

[2]All the band indices start from "0."

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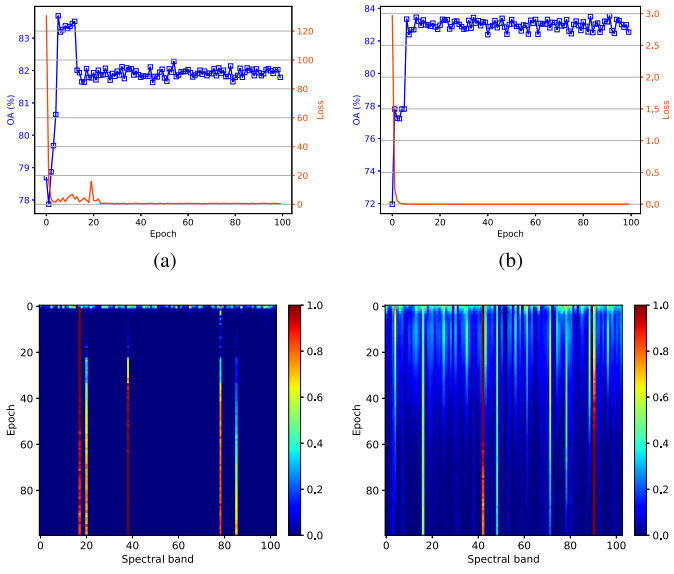
TABLE V
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS USING 25 BANDS ON THE INDIAN PINES DATA SET

No.	ISSC	SpaBS	MVPCA	SNMF	MOBS	OPBS	BS-Net-FC	BS-Net-Conv	All band
1	26.28±16.72	36.56±20.05	19.77±17.26	21.88±17.02	45.65±22.02	37.26±24.51	36.89±24.09	**59.83±28.50**	26.93±16.07
2	63.28±3.97	59.11±5.08	52.26±2.69	58.42±3.84	70.50±3.49	59.71±4.41	69.56±3.23	**74.34±3.07**	69.15±3.27
3	49.26±6.91	48.64±2.93	37.32±3.95	40.13±3.61	61.95±5.00	48.14±4.00	60.13±4.96	**62.38±4.49**	55.59±6.72
4	34.81±8.84	38.56±9.78	13.43±3.38	34.74±9.90	43.32±13.31	28.46±11.24	42.15±7.69	**54.48±14.08**	37.94±11.93
5	65.56±8.52	82.34±4.01	64.29±6.32	73.50±6.44	78.88±7.18	78.16±6.24	80.34±4.98	**83.17±5.56**	82.31±4.7
6	83.58±3.14	82.67±4.96	81.44±4.17	87.62±4.28	86.36±3.76	87.62±4.66	92.15±3.44	**94.63±2.51**	92.86±2.82
7	21.09±19.21	26.44±27.63	33.91±24.45	16.36±18.59	36.35±33.38	30.94±31.70	44.95±32.58	**52.18±40.35**	35.46±29.78
8	87.10±4.10	86.45±7.78	86.60±8.29	83.50±6.87	95.96±1.91	89.09±7.15	92.80±6.51	96.32±2.41	**96.90±2.1**
9	7.24±9.77	4.64±6.76	2.66±3.96	7.76±9.66	16.86±21.22	13.75±15.83	19.84±24.25	**28.35±26.93**	16.36±16.69
10	54.38±7.40	52.00±5.81	43.33±5.70	46.27±7.32	63.55±6.34	49.44±4.85	61.94±4.71	**69.05±5.38**	68.59±4.76
11	62.30±3.02	58.06±3.05	49.13±3.53	56.63±4.18	65.44±3.41	60.99±3.36	68.93±3.96	71.39±3.76	**77.44±2.83**
12	35.06±5.41	40.61±7.02	32.35±8.25	33.93±6.98	60.10±10.04	30.29±5.88	43.72±6.47	**64.99±9.31**	47.64±9.07
13	87.69±8.86	89.42±6.71	80.21±13.33	75.94±10.84	90.86±6.93	79.20±9.12	86.64±10.70	**95.24±2.94**	92.19±4.66
14	83.56±4.74	88.61±3.71	85.26±4.20	88.80±3.23	89.88±3.68	86.74±5.04	89.22±2.97	89.03±3.89	**93.62±2.18**
15	33.29±9.82	41.14±8.54	32.10±6.37	40.56±8.65	34.38±6.19	30.76±6.73	42.03±7.37	**49.17±7.54**	46.81±6.72
16	84.70±4.74	66.69±19.80	81.94±6.44	72.54±25.44	87.47±5.03	78.89±16.62	82.94±7.69	**85.16±5.15**	64.62±17.49
OA (%)	54.95±2.01	56.37±3.12	49.75±2.00	52.41±2.33	64.22±3.46	55.59±2.96	63.39±3.05	**70.61±2.56**	62.78±2.82
AA (%)	63.06±1.07	63.16±1.23	55.47±1.04	60.50±1.07	70.67±1.23	62.41±1.12	70.51±1.35	**75.17±1.17**	73.8±0.86
Kappa	0.579±0.012	0.580±0.014	0.494±0.011	0.5500±.012	0.666±0.014	0.572±0.013	0.664±0.015	**0.717±0.013**	0.700±0.010



(a) (b)

Fig. 8. Quantitative analysis of the band subsets selected by different BS methods on the Indian Pines data set. (a) Sum of entropy. (b) MSD values.



(a) (b) (c) (d)

Fig. 9. Analysis of the convergence of BS-Nets on the Pavia University data set. Visualization of loss versus accuracy under different iterations for (a) BS-Net-FC and (b) BS-Net-Conv. Visualization of normalized average band weights under varying iterations for (c) BS-Net-FC and (d) BS-Net-Conv.

that both BS-Net-Conv and BS-Net-FC can avoid these regions with low entropy since the noisy bands make no contribution to the spectral reconstruction. Instead, both of them select informative bands from relatively smooth regions with high entropy, thereby reducing the redundancy.

In Fig. 8(a) and (b), we further quantitatively analyze the selected band subsets by calculating the sum of entropy and MSDs of different BS methods under different band subset sizes. As can be seen from Fig. 8(a), MVPCA achieves the largest sum of entropy, but it shows the worst classification performance. This is because MVPCA selects the bands from the regions that have large entropies. However, the selected band subset contains many adjacent bands densely distributing between [150, 175], which results in high correlation. Both BS-Net-Conv and BS-Net-FC achieve competitive values in terms of total entropy when the band subset size is increased. From Fig. 8(b), BS-Net-FC achieves comparable MSD values with OPBS but is better than most of the competitors, i.e., ISSC, SpaBS, MVPCA, and MPBS. Although BS-Net-Conv achieves the best classification performance, it does not show the best MSD. As analyzed in [5], the reason for this phenomenon is that the MSD will also increase if noisy bands are selected, which can be concluded from (13). For example, the MSDs of band subsets [104, 144] and [104, 25] are 106.64 and 51.49, respectively. [104, 144] shows much better MSD than [104, 25]. Nevertheless, it contains two completely noisy bands, which makes less sense to the classification.

C. Results on Pavia University Data Set

1) Data Set: Pavia University data set was acquired by the ROSIS sensor during a flight campaign over Pavia, Northern Italy. This scene is a 103 spectral band $610 \times 6 \cdot 10$ pixels image, but some of the samples in the image contain no information and have to be discarded before the analysis. The geometric resolution is 1.3 m. The ground truth differentiates nine classes.

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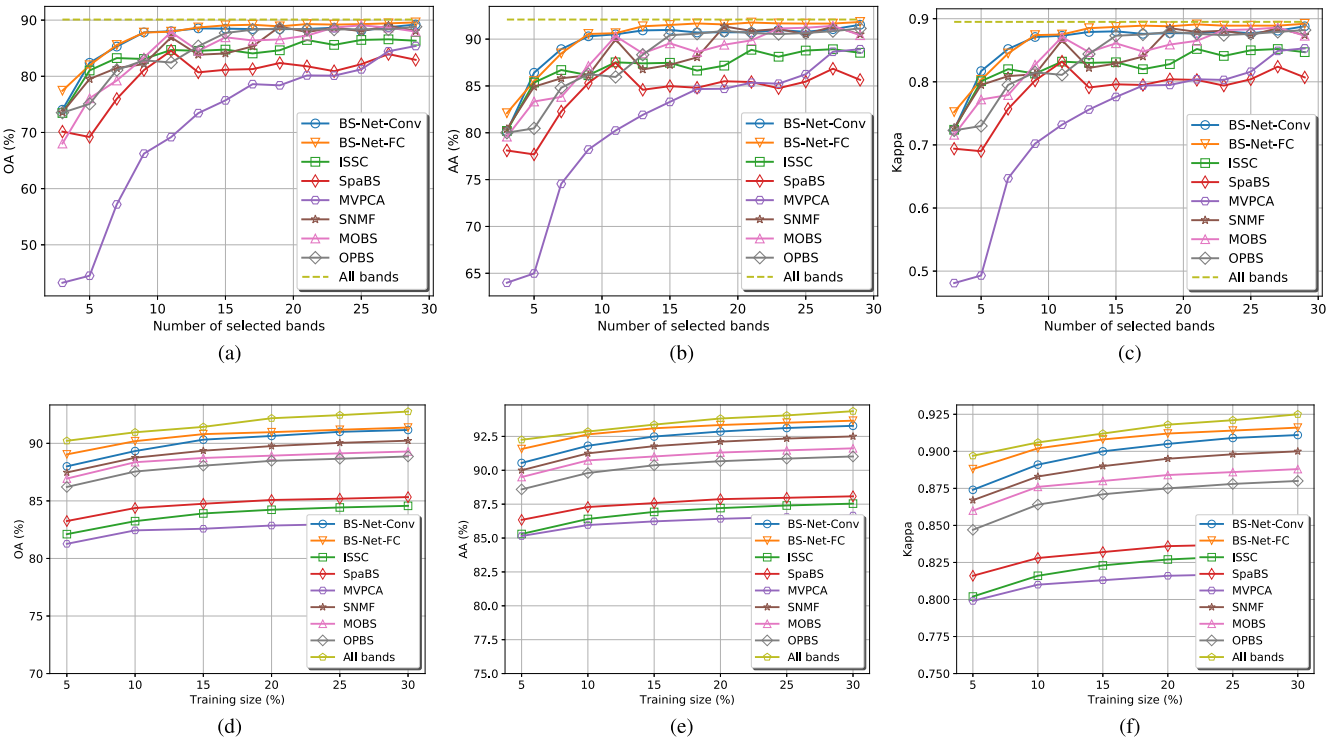


Fig. 10. Performance comparison of different BS methods with different band subset sizes (a)–(c) and different training sample sizes (d)–(e) on the Pavia University data set. (a) and (d) OA. (b) and (e) AA. (c) and (f) Kappa.

TABLE VI
PERFORMANCE COMPARISON OF DIFFERENT METHODS USING 15 BANDS ON THE PAVIA UNIVERSITY DATA SET

No.	ISSC	SpaBS	MVPCA	SNMF	MOBS	OPBS	BS-Net-FC	BS-Net-Conv	All bands
1	90.82±1.23	88.79±1.37	86.91±1.18	90.21±1.11	91.17±1.33	90.72±1.32	91.53±0.80	91.15±0.88	**91.99±1.10**
2	95.07±0.36	95.88±0.63	94.49±0.41	96.36±0.37	95.68±0.45	95.61±0.55	**96.64±0.35**	95.72±0.37	96.63±0.40
3	69.65±3.30	62.98±3.05	57.06±4.59	72.77±3.00	70.97±2.88	73.60±3.13	72.30±2.94	71.20±3.01	**76.81±2.89**
4	89.29±2.07	85.10±1.66	79.86±2.38	89.37±1.49	89.00±1.68	89.72±1.97	91.47±1.78	90.60±1.78	**93.62±1.56**
5	98.89±0.35	99.10±0.27	98.39±0.67	99.18±0.39	**99.44±0.20**	99.18±0.50	99.31±0.30	99.36±0.28	99.04±0.38
6	74.49±1.64	52.0±3.01	72.76±1.87	85.53±1.44	78.74±1.69	84.64±1.49	**86.55±1.65**	84.31±1.54	86.51±1.65
7	78.89±4.01	70.63±6.14	55.44±6.72	73.47±4.23	76.62±4.08	78.90±3.16	80.66±2.80	80.01±3.21	**83.53±3.09**
8	81.28±1.83	81.87±2.45	76.49±3.61	84.11±2.55	83.56±2.18	82.92±1.60	**85.16±2.30**	83.37±1.96	82.93±2.18
9	99.77±0.16	99.71±0.14	99.96±0.15	99.81±0.17	99.97±0.07	99.81±0.14	**99.97±0.05**	99.92±0.08	99.84±0.18
OA (%)	86.46±0.54	81.78±0.89	80.15±0.68	87.87±0.58	87.24±0.59	88.34±0.62	89.29±0.47	88.40±0.51	**90.10±0.49**
AA (%)	88.86±0.28	85.42±0.33	85.35±0.22	90.76±0.26	89.87±0.27	90.65±0.37	91.77±0.30	90.87±0.28	**92.09±0.33**
Kappa	0.852±0.004	0.803±0.005	0.804±0.003	0.877±0.003	0.865±0.004	0.876±0.005	0.891±0.004	0.879±0.004	**0.895±0.004**

2) Analysis of Convergence of BS-Nets : We show the convergence curves and the change trend of band weights in Fig. 9(a)–(d). From the results, the loss values tend to be zero after several iterations, showing that both BS-Nets converge well. Meanwhile, as the increase of iteration, the classification accuracies of the best five bands are increased from 78% to 83% and 72% to 83% for BS-Net-FC and BS-Net-Conv, respectively.

As shown in Fig. 9(c) and (d), the average band weights become very sparse and easy to distinguish when iteration increases, especially in BS-Net-FC. Finally, only a few significant bands are highlighted. These bands are useful for the spectral reconstruction. For example, one can easily determine that the significant bands of BS-Net-FC and BS-Net-Conv are [38, 78, 17, 20, 85] and [90, 42, 16, 48, 71] from Fig. 9(c) and (d), respectively.

3) Performance Comparison: In this experiment, we perform different BS methods to select different sizes of band subsets ranging from 3 to 30. We show the obtained OAs, AAs, and Kappas in Fig. 10(a)–(c), respectively. Our proposed BS-Nets have a similar classification performance and are significantly better than the other BS methods. When the band subset size is larger than 15, BS-Net-FC achieves the best classification performance, while BS-Net-Conv achieves comparable performance with SNMF and OPBS. Using more than 15 bands, both BS-Nets can achieve comparable performance with all bands. According to the VD analysis [59], [60], by setting the false alarm probability $p_F = 10^{-5}$, the best subset size of Pavia University data set is 13. However, as can be seen from Fig. 10(a)–(c), the classification performances of most of BS methods, especially BS-Nets, still increase when the band subset size is less than 15. Therefore, we use 15

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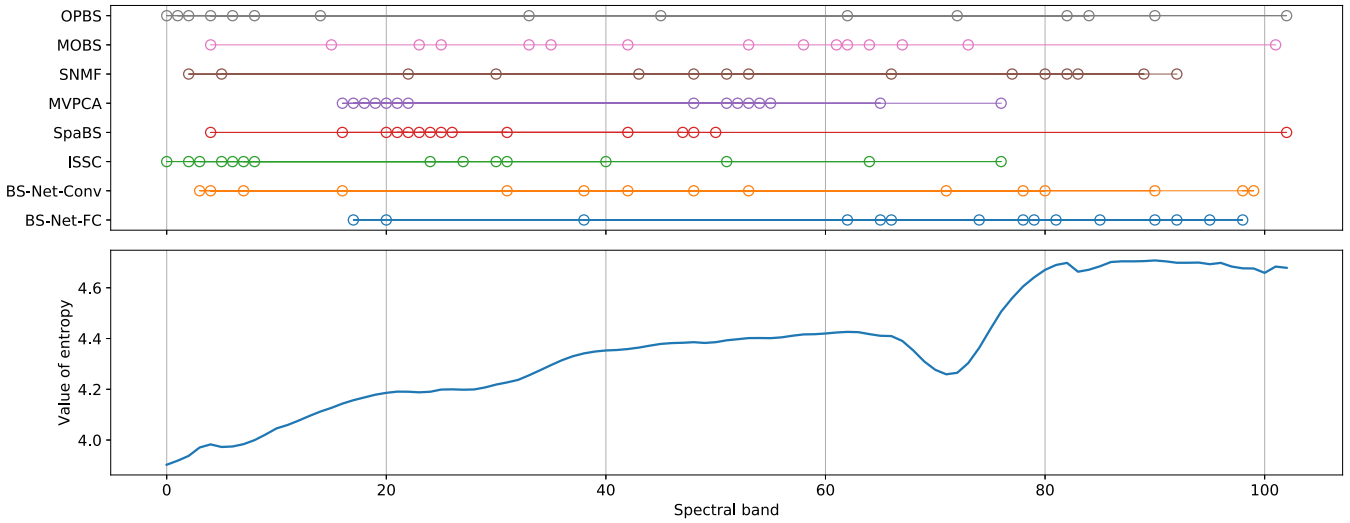


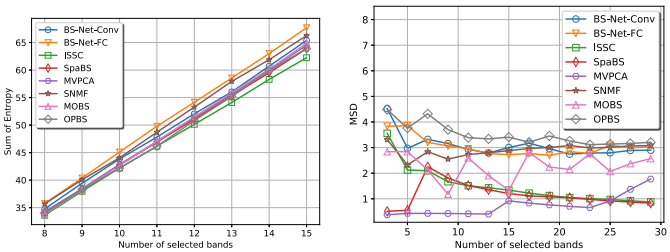
Fig. 11. Best 15 bands of Pavia University data set selected by (Top) different BS methods and (Bottom) entropy value of each band.

instead of 13 as the best subset size of Pavia University data set in the following experiments.

Fig. 10(d)–(f) shows the classification performance of using different numbers of training samples. It can be seen that BS-Net-FC and BS-Net-Conv are the best two BS methods by comparing with the other BS methods. Although all bands achieve better classification performance, our proposed BS-Nets are comparable to it. In Table VI, we compare the detailed classification performance by setting the band subset size to 15 and the training size to 5%. Similarly, BS-Net-FC and BS-Net-Conv show comparable OAs, AAs, and Kappas with all bands. For example, BS-Net-FC achieves 89.29% OA, while that achieved by all bands is 90.10%. Particularly, BS-Net-FC wins four classes, i.e., Nos. 2, 6, 8, and 9 classes. Among all the BS methods, BS-Net-FC and BS-Net-Conv are generally better than the other competitors.

4) Analysis of the Selected Bands: We extensively analyze the best 15 bands selected by different BS methods in this experiment. In Fig. 11, we show the corresponding bands distributions, as well as the entropy value of each spectral band (see the band subsets details in the Appendix). The entropy curve of Pavia University data set is relatively smooth without rapidly decreasing regions. Compared with the competitors, the band subsets selected by BS-Net-Conv and BS-Net-FC contain fewer continuous bands. Furthermore, they are concentrated around the positions with large entropy values. In contrast, MVPCA, SpaBS, and ISSC include more adjacent bands distributed over the region with low entropy. Thus, these methods achieve worse classification performance than the other BS methods.

Fig. 12(a) shows the sum of entropy of the different BS methods. According to the results, BS-Net-FC achieves the best total entropy, which demonstrates that its selected band subset consists of informative bands. SNMF shows better total entropy than BS-Net-Conv, but its classification performance is worse than BS-Net-Conv. As can be seen from Fig. 12(b), both BS-Nets show better MSDs than ISSC, SpaBS, MVPCA, and MOBS. It can be interpreted as that the band subsets selected



(a) (b)

Fig. 12. Quantitative analysis of the band subsets selected by different BS methods on the Pavia University data set. (a) Sum of entropy. (b) MSD values.

by BS-Nets contain less redundancy. Although OPBS achieves the best MSDs, its band subset includes more adjacent bands between 0 and 20. Consequently, its total entropy is lower than BS-Net-FC and BS-Net-Conv. In a nutshell, BS-Nets can accurately select the significant band subsets with guarantees of entropy values and MSD values.

D. Results on Salinas Data Set

1) Data Set: This scene was collected by the 224-band AVIRIS sensor over Salinas Valley, CA, USA, and is characterized by high spatial resolution (3.7-m pixels). The area covered comprises 512 × 217 samples. As with Indian Pines scene, we discarded the 20 water absorption bands, in this case, bands: [108 − 112], [154 − 167], and 224. It includes vegetables, bare soils, and vineyard fields. Salina's ground truth contains 16 classes.

2) Analysis of Convergence of BS-Nets: Fig. 13(a) and (b) shows the convergence curves of BS-Nets on the Salinas data set. Training about 20 iterations, BS-Nets' loss and accuracy have tended to be convergent. The OA of using five bands is increased from 92% to 94% and from 85% to 94% for BS-Net-FC and BS-Net-Conv, respectively. In Fig. 13(c) and (d), we show the means of band weights under different iterations. Similar to the Indian Pines and Pavia University data sets,

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TABLE VII
PERFORMANCE COMPARISON OF DIFFERENT METHODS USING 20 BANDS ON THE SALINAS DATA SET

NO.	ISSC	SpaBS	MVPCA	SNMF	MOBS	OPBS	BS-Net-FC	BS-Net-Conv	All bands
1	98.72±0.86	98.40±0.85	97.95±1.17	99.30±0.24	98.69±0.72	98.76±0.68	99.32±0.55	**99.39±0.31**	99.14±0.42
2	99.70±0.29	99.40±0.35	97.85±0.79	99.71±0.16	99.53±0.25	99.69±0.26	99.58±0.37	**99.72±0.25**	99.66±0.21
3	97.66±0.92	97.83±1.15	90.07±2.73	97.61±0.99	99.08±0.57	97.20±1.28	99.30±0.37	99.25±0.37	**99.40±0.28**
4	98.94±0.74	98.80±0.78	98.95±0.77	98.79±1.30	99.14±0.65	98.76±0.82	98.99±0.61	98.64±1.01	**99.33±0.33**
5	98.27±0.51	97.77±1.07	94.60±1.10	97.61±0.80	97.98±0.62	96.96±1.14	97.98±0.80	98.39±0.65	**98.51±0.64**
6	**99.85±0.06**	99.69±0.17	99.12±0.48	99.84±0.08	99.77±0.09	99.79±0.09	99.79±0.11	99.79±0.12	99.79±0.17
7	**99.64±0.25**	99.45±0.42	98.72±0.56	99.52±0.27	99.59±0.18	99.56±0.22	99.58±0.23	99.56±0.15	99.45±0.22
8	84.33±1.31	84.87±1.94	83.77±2.17	85.71±1.36	86.24±1.23	85.06±1.71	87.27±1.60	**88.15±1.13**	81.65±1.54
9	99.30±0.46	99.24±0.52	93.84±0.76	98.99±0.73	98.94±0.62	99.06±0.73	99.34±0.38	99.46±0.42	**99.79±0.13**
10	91.49±1.27	90.27±1.13	84.31±1.71	92.93±0.99	93.73±1.25	91.75±1.75	94.95±1.12	94.82±1.01	**95.20±1.18**
11	91.72±2.13	92.41±2.90	67.54±5.53	94.53±2.56	94.21±1.77	95.33±2.91	94.09±3.34	96.19±1.51	**97.16±1.89**
12	97.92±0.85	98.74±0.58	86.78±2.61	98.98±0.72	99.58±0.33	99.47±0.54	99.49±0.88	99.59±0.71	**99.70±0.26**
13	98.07±1.28	97.63±1.19	96.43±1.82	98.08±1.02	98.09±1.34	97.57±1.85	98.72±1.25	**98.94±0.85**	98.06±1.12
14	94.64±2.11	93.96±2.83	94.63±1.71	94.27±2.47	95.58±1.08	94.84±1.99	96.18±1.55	**96.76±1.53**	94.51±3.77
15	61.18±2.42	65.22±2.55	52.00±3.13	65.17±1.79	69.83±1.92	70.24±2.60	71.60±2.34	70.41±2.23	**73.62±2.27**
16	97.60±0.86	97.63±0.64	93.99±1.64	98.48±0.51	97.76±0.56	**98.88±0.21**	98.08±0.82	98.65±0.58	98.52±0.46
OA (%)	94.31±0.18	94.46±0.27	89.41±0.37	94.97±0.21	95.48±0.18	95.18±0.34	95.89±0.31	**96.11±0.17**	95.84±0.26
AA (%)	90.18±0.22	90.69±0.21	86.01±0.25	91.14±0.18	91.97±0.20	91.60±0.20	92.61±0.19	**92.74±0.20**	91.84±0.36
Kappa	0.890±0.002	0.896±0.002	0.844±0.003	0.901±0.002	0.911±0.002	0.906±0.002	0.918±0.002	**0.919±0.002**	0.909±0.004

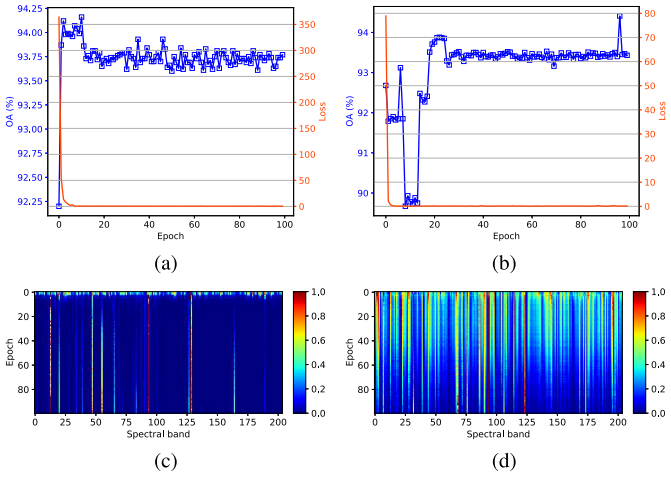


Fig. 13. Analysis of the convergence of BS-Nets on the Salinas data set. Loss versus accuracy under different iterations for (a) BS-Net-FC and (b) BS-Net-Conv. Visualization of normalized average band weights under varying iterations for (c) BS-Net-FC and (d) BS-Net-Conv.

the learned band weights become sparse with the increase of iterations.

3) Performance Comparison: In this section, we compare the classification performance of different BS methods on the Salinas data set. First, we analyze the classification performance with different band subset sizes in Fig. 14(a)–(c). From the results, BS-Net-FC and BS-Net-Conv can achieve better classification performance than ISSC, SpaBS, MVPCA, SNMF, and OPBS, as well as all bands. When the band subset size is less than 20, BS-Net-FC achieves the best classification performance, whereas using more than 20 bands, BS-Net-Conv shows slightly better performance. BS-Net-FC can achieve better OA, AA, and Kappa than all bands by using 15, 10, and 10 bands, respectively. When the band subset size is larger than 20, the classification performances of most of the BS methods are no longer increased. Interestingly, this subset size is consistent with that estimated by the VD analysis [60].

TABLE VIII
COMPUTATIONAL TIME (IN SECONDS) OF USING DIFFERENT BS METHODS TO SELECT 19 BANDS

Method	Indian Pines	Pavia University	Salinas
ISSC	0.43	14.71	17.44
SpaBS	332.54	2026.80	3224.57
MVPCA	0.44	4.24	7.86
SNMF	$> 1h$	$> 1h$	$> 1h$
MOBS	321.64	487.91	578.20
OPBS	38.22	65.60	71.78
BS-Net-FC	741.12	2907.80	1788.17
BS-Net-Conv	319.54	531.88	503.39

For the sake of fairness, we will set the band subset size to be 20 for all the BS methods in the following tests.

Second, we evaluate the classification performance with the different number of training samples for different BS methods. The results are shown in Fig. 14(d)–(f). Although the classification performance of all the BS methods can increase as the training size increased, BS-Net-Conv is better than the other BS methods. Especially, when the training size is less than 10%, BS-Net-Conv shows a tendency to outperform all bands. Additionally, BS-Net-FC's classification performance is comparable with MOBS and OPBS. The detailed classification results using 20 bands and 5% training samples are given in Table VII. It can be seen that BS-Net-Conv shows the best classification results in terms of OA, AA, and Kappa. Furthermore, for Nos. 1, 2, 8, 13, and 14 classes, BS-Net-Conv achieves the best accuracy. BS-Net-FC's performance is comparable with BS-Net-Conv and better than the other BS methods, as well as all bands.

4) Analysis of the Selected Bands: To further analyze the selected band subsets, we show their distribution and entropy of each band in Fig. 15. As can be seen, BS-Net-FC and BS-Net-Conv contain fewer adjacent bands and distribute relatively uniformly. Observing from the entropy curve, BS-Net-FC and BS-Net-Conv can avoid the sharply decreasing regions with low entropy, i.e., [106, 107] and [146, 147].

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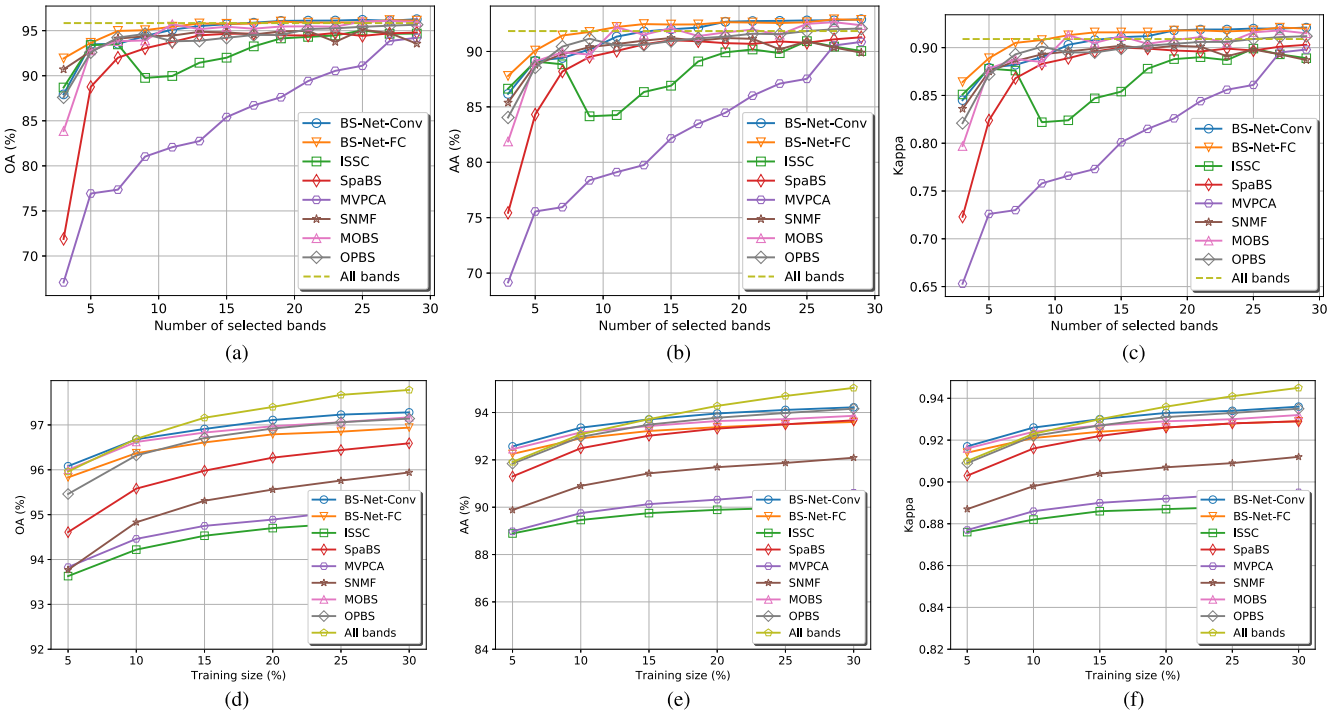


Fig. 14. Performance comparison of different BS methods with different band subset sizes (a)–(c) and different training sample sizes (d) and (e) on the Salinas data set. (a) and (d) OA. (b) and (e) AA. (c) and (f) Kappa.

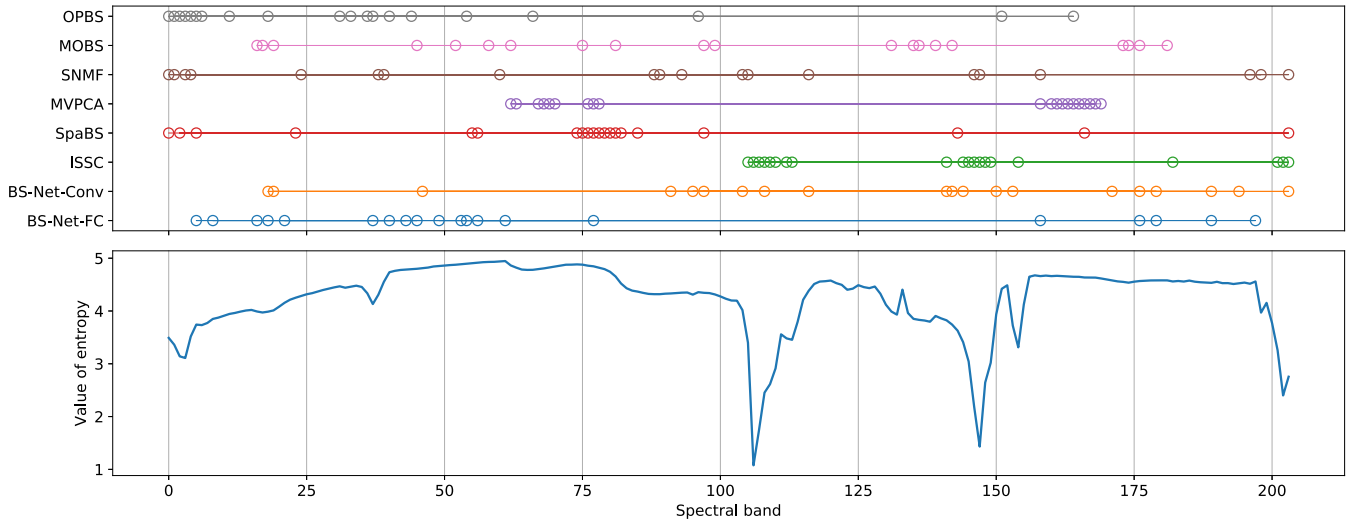


Fig. 15. Best 20 bands of the Salinas data set selected by (Top) different BS methods and (Bottom) entropy value of each band.

From Fig. 15 (top), some selected band subsets consist of many continuous bands, such as OPBS, MVPCA, SpaBS, and ISSC. Therefore, higher correlations are contained in these band subsets.

The total entropy and MSDs of the selected band subsets are shown in Fig. 16(a) and (b), respectively. Similar to the Indian Pines data set, MVPCA achieves the best total entropy on the Salinas data set. However, its bands' distribution is dense and the MSDs are the lowest. In other words, its selected bands are highly correlated and include the most redundancy. By contrast, BS-Net-Conv achieves comparable total entropy with MVPCA. Furthermore, its classification performance is the best among the BS methods. As shown in Fig. 16(b), BS-Net-Conv achieves better MSDs than SpaBS, MVPCA, and OPBS. Although ISSC obtains better MSDs than BS-Net-Conv, its total entropy is lowest due to the selection of bands from the noisy regions. BS-Net-FC shows relatively lower MSDs, but it achieves better classification performance than most of the BS methods. To sum up, BS-Net-Conv is superior to the other BS methods in terms of classification performance and quantitative indices.

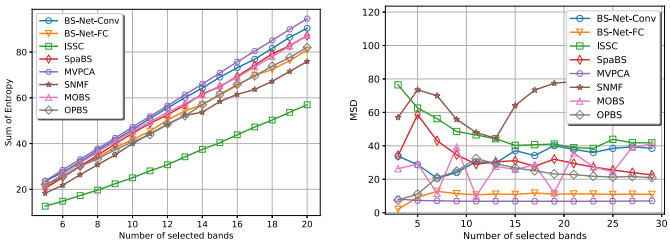
E. Computational Time Complexity Analysis

In this section, we compare the running time of different BS methods. To conduct a fair experiment, all the BS methods are

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TABLE IX
BEST BAND INDICES OF INDIAN PINES, PAVIA UNIVERSITY, AND SALINAS DATA SETS SELECTED BY DIFFERENT BS METHODS

Data	Methods	Selected Bands
InP.	BS-Net-FC	[165, 38, 51, 65, 12, 100, 0, 71, 5, 60, 88, 26, 164, 75, 74, 52, 22, 94, 35, 11, 184, 179, 34, 160, 46]
	BS-Net-Conv	[46, 33, 140, 161, 80, 35, 178, 44, 126, 36, 138, 71, 180, 66, 192, 16, 53, 152, 185, 119, 24,28,26,156,83]
	ISSC	[171, 130, 67, 85, 182, 183, 47, 143, 138, 90, 139, 141, 25, 142, 21, 181, 3, 34,13,14,11,10,190,187,189]
	SpaBS	[7, 96, 52, 171, 53, 3, 76, 75, 74, 95, 77, 73, 78, 54, 81, 94, 88, 91, 71, 72, 79, 80, 55, 92, 56]
	MVPCA	[167,74,168,0,147,165,161,162,152,19,160,119,164,159,157,163,158,156,20,154,118,148,153,149,155]
	SNMF	[23, 197, 198, 94, 76, 2, 87, 105, 143, 145, 11, 84, 132, 108, 28, 104, 144, 34, 44, 74, 71, 96,75,171,162]
	MOBS	[8, 14, 21, 22, 30, 33, 39, 51, 52, 64, 79, 80, 86, 87, 106, 114, 124, 128, 135, 141, 160, 168,169,182,196]
	OPBS	[28, 41, 60, 0, 74, 34, 88, 19, 17, 33, 56, 87, 22, 31, 73, 12, 18, 32, 90, 39, 11, 89, 75, 24, 70]
PaU.	BS-Net-FC	[38, 78, 17, 20, 85, 98, 65, 81, 79, 90, 95, 74, 66, 62, 92]
	BS-Net-Conv	[90, 42, 16, 48, 71, 3, 78, 38, 80, 53, 7, 31, 4, 99, 98]
	ISSC	[51, 76, 7, 64, 31, 8, 0, 24, 40, 30, 5, 3, 6, 27, 2]
	SpaBS	[50, 48, 16, 22, 4, 102, 21, 25, 23, 47, 24, 20, 31, 26, 42]
	MVPCA	[48, 22, 51, 16, 52, 21, 65, 17, 20, 53, 18, 54, 19, 55, 76]
	SNMF	[92, 53, 43, 66, 22, 89, 82, 30, 51, 5, 83, 77, 80, 2, 48]
	MOBS	[4, 15, 23, 25, 33, 35, 42, 53, 58, 61, 62, 64, 67, 73, 101]
	OPBS	[90, 62, 14, 0, 2, 72, 102, 4, 33, 1, 6, 84, 45, 82, 8]
Sal.	BS-Net-FC	[53, 77, 61, 54, 16, 8, 158, 49, 176, 179, 56, 189, 197, 21, 43, 37, 5, 40, 45, 18]
	BS-Net-Conv	[116, 153, 19, 189, 97, 179, 171, 141, 95, 144, 142, 46, 104, 203, 91, 18, 176, 108, 150, 194]
	ISSC	[141, 182, 106, 147, 107, 146, 108, 202, 203, 109, 145, 148, 112, 201, 110, 113, 144, 149, 105, 154]
	SpaBS	[0, 79, 166, 80, 203, 78, 77, 76, 55, 81, 97, 5, 23, 75, 2, 82, 56, 74, 143, 85]
	MVPCA	[169, 67, 168, 63, 68, 78, 167, 166, 165, 69, 164, 163, 77, 162, 70, 62, 160, 161, 76, 158]
	SNMF	[24, 1, 105, 196, 203, 0, 39, 116, 38, 60, 89, 104, 198, 147, 158, 3, 146, 4, 93, 88]
	MOBS	[16, 17, 19, 45, 52, 58, 62, 75, 81, 97, 99, 131, 135, 136, 139, 142, 173, 174, 176, 181]
	OPBS	[44, 31, 37, 66, 11, 1, 164, 2, 18, 0, 3, 40, 4, 54, 33, 96, 5, 36, 6, 151]



(a) (b)

Fig. 16. Quantitative analysis of the band subsets selected by different BS methods on the Salinas data set. (a) Sum of entropy. (b) MSD values.

implemented in Python and executed on the same CPU platform. Table VIII shows the results of selecting 19 bands using different BS methods on the three data sets. As can be seen, the running times of BS-Net-Conv are significantly less than SpaBS and SNMF and more competitive than MOBS. Since ISSC and MVPCA can be solved using algebraic methods, they are faster than the other BS methods. Nevertheless, neither of them achieves better performance than BS-Nets. For all the three data sets, BS-Net-Conv is generally faster than BS-Net-FC. According to the implementation details shown in Table III-IV, BS-Net-FC includes 152 592 trainable parameters and 21 025 training samples, whereas that involved in BS-Net-Conv are 590 288 and 4 489, respectively. It can be seen that BS-Net-Conv uses fewer training samples to train more parameters and saves more training time. For the other BS methods, e.g., ISSC, their numbers of samples are equal to the total number of bands, which are far less than that used in BS-Nets. Furthermore, it is noticeable that BS-Nets are easily accelerated with GPU. In summary, the proposed BS-Nets are able to balance classification performance and running time.

V. CONCLUSION

This article presents a novel end-to-end BS-Net framework for HSI BS. The main idea behind the framework is to treat HSI BS as a sparse spectral reconstruction task and to explicitly learn the spectral band's significance using DNNs by considering the nonlinear correlation between the spectral bands. The resulting framework allows to learn band weights from full spectral bands, resulting in more efficient use of the global spectral relationship, and consists of two flexible subnetworks, BAM and RecNet, making it easy to train and apply in practice. The experimental results show that the implemented BS-Net-FC and BS-Net-Conv can not only adaptively produce sparse band weights but also significantly better classification performance than many existing BS methods with an acceptable time cost. Since the spatial information is used in BS-Net-Conv, BS-Net-Conv generally achieves better performance than BS-Net-FC.

We notice that the proposed framework has the capacity of combining with many deep learning-based classification methods to reduce computational complexity and enhance the classification performance, which will also be further explored in our future works.

APPENDIX

In this section, we show the selected band indices determined by different BS methods for Indian Pines, Pavia University, and Salinas data sets in Table IX. The numbers of selected bands of the three data sets are 25, 15, and 20, respectively. For the convenience of Python coding, the band indices start from " 0," e.g., "165" means that the 166th band is selected.

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